

AGENT MEMORY INFRASTRUCTURE

The TDAI Four-Layer Memory System

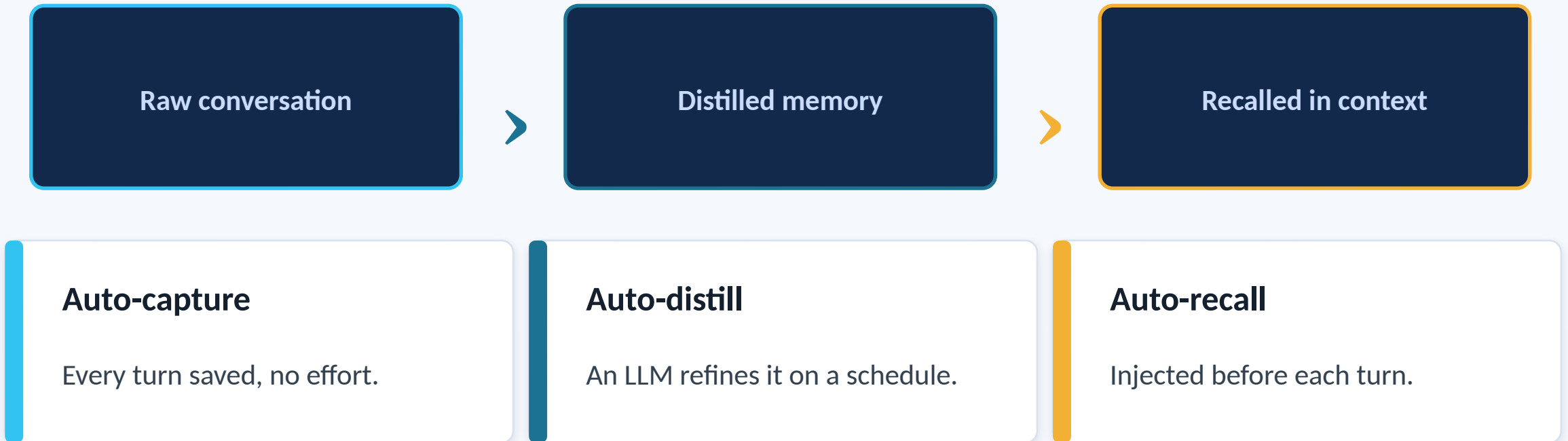
What it is • How agents use it • Hosting 1,000 agents

Prepared for Rick Stevens • Kukla • June 2026

OVERVIEW

What the system is

It turns raw agent conversations into durable, searchable knowledge — automatically.



Why agents need this

The problem

Stateless agents wake up blank. Context is finite.
Monday's knowledge is gone by Tuesday.

The naive fix

Dump it all in the prompt. Blows the budget, drowns the signal.

The TDAI answer

A tiered hierarchy — like cache → RAM → disk.

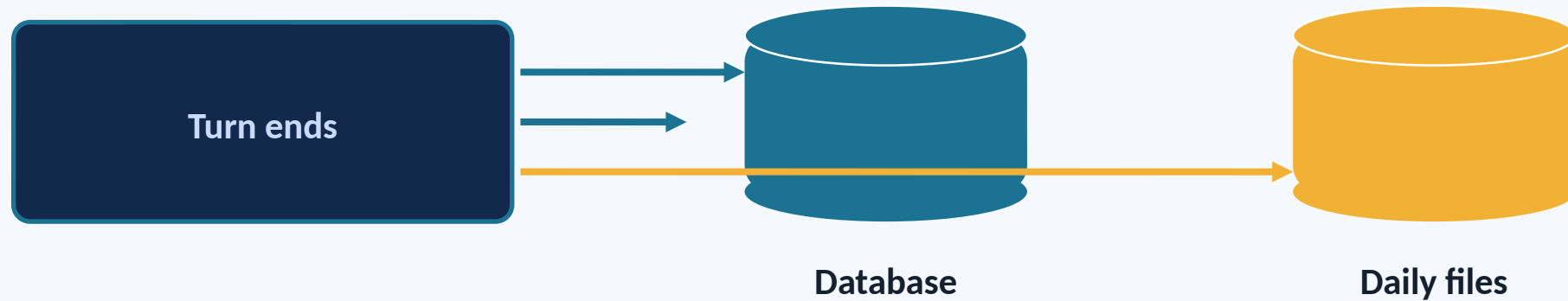
Hot memory is injected. The long tail stays searchable on demand.

The agent carries a compact working set, not its whole past.

The four layers



Lo — Conversation recording



Dual-write = durability

Database for speed, flat files for portability and recovery. Either rebuilds the other.

On this machine

496 raw turns recorded and full-text indexed for instant search.

L1 — Memory extraction

Every few conversations, an LLM distills durable facts into three types:

persona

Stable traits & preferences.

“Wants LLM judgment, not regex.”

episodic

Events, with dates.

“Fixed the cap bug on 06-21.”

instruction

Standing rules.

“Never echo passwords.”

Each memory has a priority. Duplicates are detected and merged automatically.

L2 scenes → L3 persona

L2 — Scene Blocks

Related memories grouped into a readable narrative chunk.

“Helping Rick diagnose model-serving failures.”

A middle unit between one fact and the whole history.

L3 — Persona

From all scenes, an LLM synthesizes one profile, updated continuously.

The most-compressed layer — injected first on every recall.

The agent's standing model of its human.

MECHANICS

The promotion pipeline

Layers fire in sequence, in the background — the conversation never waits.



Capture is instant; distillation is background work — the agent is never slowed.

How memory is searched

Keyword

Exact term match. Fast, transparent.

Semantic

Meaning-based. Related ideas, different words.

Hybrid

Blends both. The default.

Two tools the agent can call

`tdai_memory_search` — facts, preferences, rules (L1)

`tdai_conversation_search` — exact past dialogue, timelines (L0)

ISOLATION

One engine, many agents

Each agent gets its own isolated store under one gateway. Six are live today:



kukla



ollie-test



hermes



piago



piago-test



smoke-test

Built for scale already

Store pool sized to 1,000 • shared search encoder • 16-way parallel scheduler.

ON THIS MACHINE

The system, live

496

raw turns

150

memories

6

agents

<5ms

recall

What this means

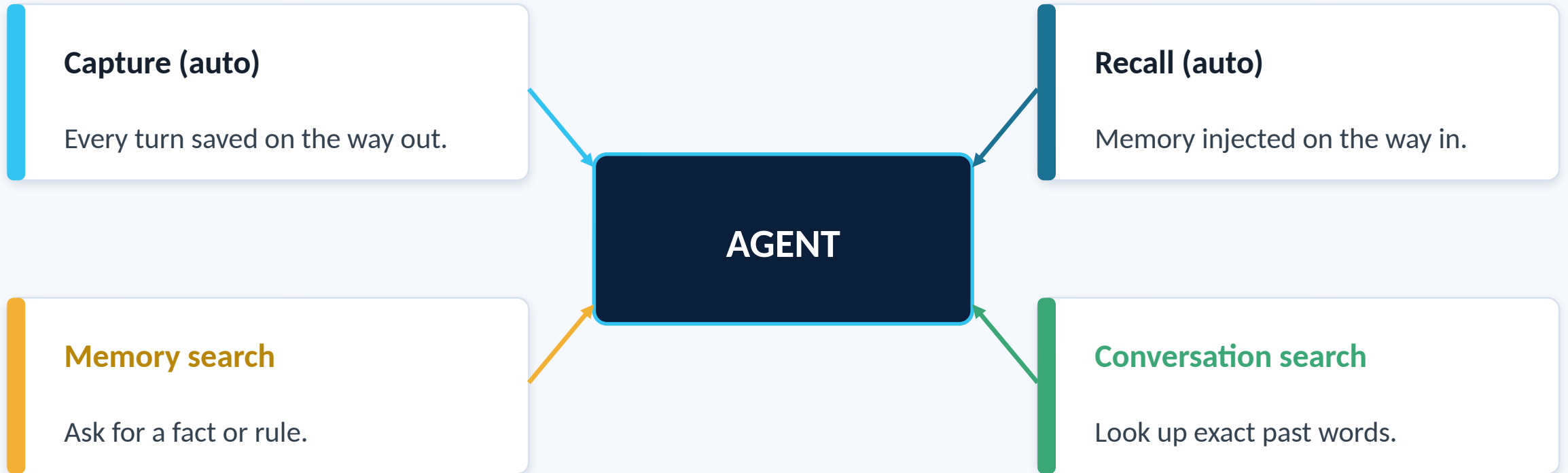
From 496 captured turns, the pipeline distilled 150 structured memories — serving sub-5-ms searches across 6 isolated agents, all on one Mac mini.

PART TWO

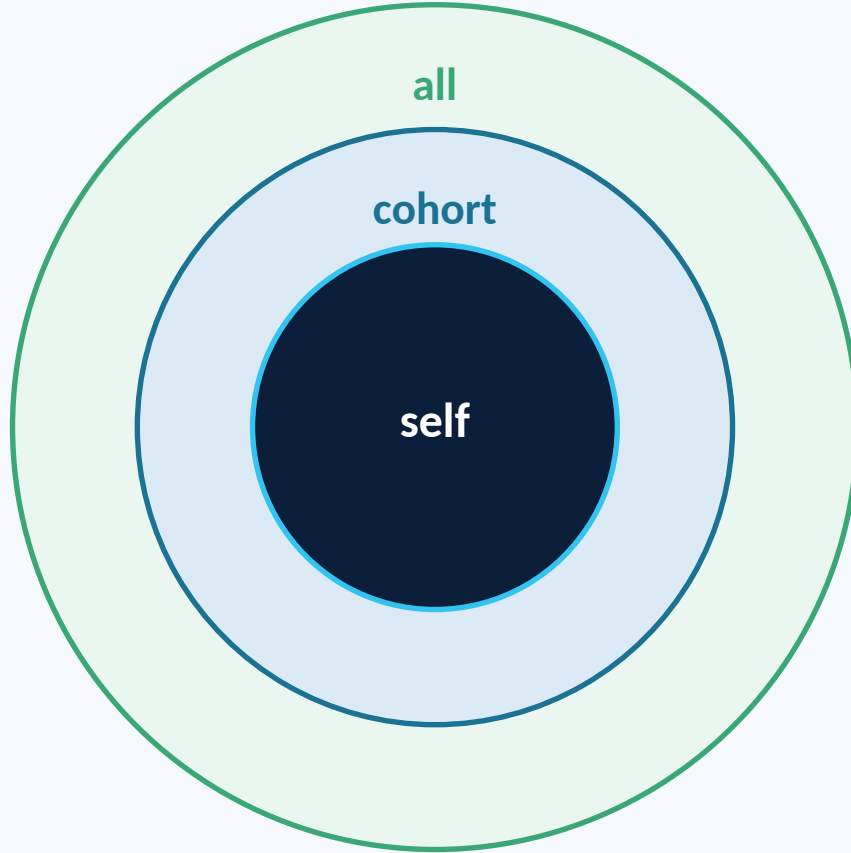
How agents use it

Four operations + a query scope

Four operations



Query scope



self

Only this agent's memory. The private default.

cohort

A named group of agents' shared knowledge.

all

The collective memory of the whole fleet.

PART THREE

Hosting 1,000 agents

Three tiers that scale independently

TARGET ARCHITECTURE

Three tiers



Agents never block on each other; the queue absorbs load and preserves order.

Five phases

1

Decouple capture

Fast queue in front; front-end goes stateless.

2

Externalize storage

Single-file → clustered store. One adapter.

3

Distribute pipeline

Scheduler goes multi-node; autoscale extraction.

4

Build aggregation

The new cross-agent merge + fleet search.

5

Load-test to 1,000

Prove the targets under full load.

SUMMARY

Bottom line

What it is

A local, four-layer memory engine.

How it's used

Four operations; scoped self / cohort / all.

Scaling it

Three independent tiers; the bones exist.

The hard part

LLM-extraction throughput — not storage.